Maulana Azad National Institute of Technology Bhopal

End-Term Examination (March 2022)

Course: B.Tech. **Semester-I** **Section: All**

Subject: Mathematics-I **Subject Code: MTH-101**

Time:3:00 Hours **Max. Marks: 50**

Note: All questions are compulsory.

S.No.	Question	Marks
1	Expand *cos x* using Taylor series upto order 4 about the point x = π/6. Can we calculate the value of *cos 32°* using Taylor Series? If yes, calculate the value of cos 32°.	05
2(a)	If $z = f(u,v), u = e^x \cos y, v = e^x \sin y$, show that $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = (u^2+v^2)\left(\frac{\partial^2 z}{\partial u^2} + \frac{\partial^2 z}{\partial v^2}\right)$	05
2(b)	Using the properties of beta and gamma functions, show that $\int_0^{\pi/2} \sqrt{\sin\theta}\, d\theta \times \int_0^{\pi/2} \frac{1}{\sqrt{\sin\theta}}\, d\theta = \pi$	05
3(a)	Change the order of integration and hence evaluate $\int_0^a \int_{\sqrt{ax}}^a \frac{y^2\, dydx}{\sqrt{y^4 - a^2x^2}}$	05
3(b)	Find the area of a loop of the curve $a^2x^2 = 4y^2(a^2 - y^2)$	05
4(a)	Verify the Green's theorem for $f(x,y) = e^{-x}\sin y, g(x,y) = e^{-x}\cos y$, where C is the square with vertices at (0,0), (π/2,0), (π/2, π/2) and (0, π/2).	05
4(b)	Calculate $\iint_S \vec{F}.\vec{n}\, dS$, where $\vec{F} = 4xz\,\vec{i} - y^2\vec{j} + yz\,\vec{k}$ and S is the surface of the cube bounded by the planes x = 0, x = 2, y = 0, y = 2, z = 0, z = 2.	05
5(a)	Solve the system of differential equations $\frac{d^2x}{dt^2} - \frac{dy}{dt} = 2x + 2t$, $\frac{dx}{dt} + 4\frac{dy}{dt} = 3y$	08
5(b)	Solve the following differential equation by using the method of variation of parameter $(3x+2)^2 \frac{d^2y}{dx^2} + 3(3x+2)\frac{dy}{dx} - 36y = 3x^2 + 4x + 1$	07

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Maulana Azad National Institute of Technology Bhopal-462003

End Term Examination (Session: 2022-23)

Course: B.Tech. **Semester-I** **Section: All** **Time: 3 Hours**

Subject: Mathematics-I **Subject Code: MTH-101** **Max. Marks: 50**

Note: All questions are compulsory.

S.No.	Question	Marks
1(a)	If $\theta = f\left(\frac{y-x}{xy}, \frac{z-x}{xz}\right)$, *show that* $x^2\frac{\partial\theta}{\partial x} + y^2\frac{\partial\theta}{\partial y} + z^2\frac{\partial\theta}{\partial z} = 0$	4
1(b)	Find an approximate value of f(1. , 0.8) using the Taylor series quadratic approximation, if $f(x,y) = \tan^{-1}(xy)$.	4
2(a)	Discuss the maxima and minima of the function $u = \cos A \cos B \cos C$ where A, B, C are the angles of a triangle.	4
2(b)	If $u = \sin^{-1}\frac{x+2y+3z}{x^8+y^8+z^8}$, find the value of $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z}$.	4
3(a)	Find the area lying inside the circle $r = \sin\theta$ and outside the cardioid $r = (1-\cos\theta)$.	5
3(b)	The loop of the curve $2ay^2 = x(x-a)^2$ revolves about the x-axis. Find the volume of the solid so generated.	5
4(a)	Verify the Stokes's theorem for the vector field $v = (3x-y)i - 2yz^2 j - 2y^2 z k$, where S is the surface of the sphere $x^2 + y^2 + z^2 = 16, z > 0$	4
4(b)	Use the divergence theorem to evaluate $\iint_S (V.n)\,dA$, *where* $V = x^2 z i + y j - xz^2 k$ and S is the boundary of the region bounded by the paraboloid $x^2 + y^2 = z$ and the plane $z = 4y$.	4
4(c)	Evaluate $\oint_C (x^2 + y^2)dx + (y + 2x)dy$, where C is the boundary of the region in the first quadrant, which is bounded by the curves $x^2 = y$ & $y^2 = x$.	4
5(a)	Solve $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 8(x^2 + e^{2x} + \sin 2x)$	4
5(b)	Solve by using the method of variation of parameter $x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} - y = x^2 e^x$	4
5(c)	Solve $(y - \frac{1}{3}y^3 + \frac{1}{2}x^2)dx + \frac{1}{4}(x + xy^2)dy = 0$	4

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Scholar No. :

Maulana Azad National Institute of Technology, Bhopal-462003
End Term Examination (Session: July-December 2024)

B.Tech. Semester - I Section: All

Subject: Mathematics-I Subject Code: MTH-24101

Max. Marks: 40 Time: 120 Minutes

Note:
- All questions are compulsory. Usual notations are used.
- Assume missing data if any.

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1. Discuss the maximum and minimum values of the given function [6 marks]

$$u = a(x^3y^2 - x^4y^2 - x^3y^3)$$

2. Trace the curve $r^2 = a^2 \sin 2\theta$ by describing the necessary properties. [6 marks]

3. Evaluate the double integral $\iint_R \sqrt{x^2+y^2}\,dxdy$, where R is the region in the xy plane bounded by the circles $x^2+y^2=4$ and $x^2+y^2=9$. [6 marks]

4. Verify Stokes theorem for the surface S given by $x^2+y^2+z^2=9, z \geq 0$ and the vector field $\vec{v} = y\hat{i} - x\hat{j}$. [6 marks]

5. Find the work done by the force $\vec{F} = (x^2-y^3)\hat{i} + (x+y)\hat{j}$ in moving a particle along the closed path C containing the curves $x+y=0$, $x^2+y^2=16$ and $y=x$ in the first and fourth quadrants. [5 marks]

6. Find the general solution of the given linear differential equation [6 marks]

$$\frac{d^4y}{dx^4} + 2\frac{d^3y}{dx^3} - 3\frac{d^2y}{dx^2} = x^2 + 3e^{2x} + 4\sin x$$

7. Solve the differential equation satisfying the initial condition $y(1)=2$ [5 marks]

$$(x^2+y^2)dx - 2xydy = 0$$

** End of the question paper **

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